Bhavin K. Moriya

Data Scientist | Machine Learning | Privacy-Preserving Analytics
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Google Scholar GitHub LinkedIn Kaggle YouTube RoutingApp StreamlitApps

Experience

Data Scientist / Research Assistant **Hochschule Esslingen, Germany** **Nov 2023 – Dec 2025**

- **AnoMoB Project:** Developed first-of-its-kind encrypted comparison operations and non-linear function evaluations using **OpenFHE (C++)** and **TFHE-rs (Rust)**.
- **Sustainability Analysis:** Leveraged **SUMO** to simulate movement data and conducted comparative analyses of travel modalities focusing on **CO2 emissions** and optimal route duration.
- **Performance Benchmarking:** Co-authored breakthrough research comparing runtime and memory consumption of **CKKS and TFHE** schemes, published in Cryptology ePrint Archive (2025).
- **Data Engineering:** Built production-quality software tools for statistical analysis of mobility data using **Python, polars, and geopandas**.

Assistant Professor **Univ. of Viçosa, Brazil** **Nov 2015 – Nov 2023**

- Taught undergraduate and master's level mathematics; supervised 3 dissertations.
- Collaborated with international researchers, resulting in 16+ peer-reviewed publications.

Postdoctoral Fellowships

- IMSc (2011–2013), HRI (2013–2014), CMI (2014), Univ. of Brasília (2014–2015)

Education

- **Ph.D. in Mathematics**, Harish Chandra Research Institute **2011**
  Thesis: Some Zero Sum Problems in Combinatorial Number Theory
- **M.Sc. & B.Sc. in Mathematics**, The M.S. University of Baroda **2003 – 2005**

Skills

Programming: Python (Proficient), C++, Rust, SQL, LaTeX, Go
Data & Simulation: polars, geopandas, SUMO, Scikit-learn, Docker
Workflow: LLM-assisted development (GitHub Copilot), Git, **Agile Research**, **Streamlit**
Languages: English (Fluent), German (**B1/Professional Communication**), Portuguese (Fluent/Professional), Hindi (Native), Gujarati (Native)

Selected Projects (GitHub)

- **Energy Consumption Forecasting:** Developed tools for energy consumption forecasting and modeling.
- **Quantitative Finance Modeling:** Implementation of financial modeling and algorithmic trading concepts.

Publications & Awards

- (2025) **A Performance Comparison of Homomorphic Encryption Schemes CKKS and TFHE**, Cryptology ePrint Archive.
- 16 peer-reviewed publications in combinatorics and cryptography.
- Recipient of multiple international grants (ICM, CIMPA-ICTP, FAPEMIG).